CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College of Ormoc

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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May 12, 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City**

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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This capstone project proposal titled **CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City,** prepared and submitted by **Steve Clarence D. Amores, Czar Aian L. Judilla**, **Nick Andre N. Suson,** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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TABLE OF CONTENTS

	Page
Title Page	i
Endorsement form for Proposal Defense	ii
Approval Sheet	iii
Table of Contents	iv
Introduction	1
Project Context	1-2
Purpose and Description	3
Objective	4-6
Scope and Limitations	7-12
Review of Related Literature/Studies/Systems	12-16
Methodology	17
Technical Background	17-19
Requirements Analysis	20
Requirements Documentation	21-22
Design of Software, System, Product, and/or Processes	23-24
References	25-26
Appendices	27
Resource Persons	28-29
Personal Technical Vitae	30-33
Design Model	34-41

INTRODUCTION

Project Context

Tourism is one of the key sectors that contributes to local economic growth and cultural promotion. In Ormoc City, various natural, cultural, historical, and recreational attractions continue to attract both local and foreign visitors. As tourism activities increase, there is also a growing need for efficient access to tourism-related information and services that can help tourists plan their trips and enhance their overall travel experience.

Despite the availability of tourism destinations in Ormoc City, tourists often encounter difficulties in gathering accurate and organized information about places they wish to visit. Information is commonly scattered across social media pages, websites, brochures, and word-of-mouth recommendations. As a result, tourists spend considerable time searching for destinations, comparing options, and organizing their travel plans, which can lead to inconvenience and inefficient trip preparation.

Ormoc City already has an existing tourism website that provides information about tourist destinations and tourism-related updates. However, the current platform mainly serves as an informational website with limited functionality. While users can view details about tourist spots, the system does not provide interactive services that support complete travel planning and guided experiences. It lacks features such as itinerary creation, and tour guide booking, which are essential for a more complete and modern travel experience.

Another challenge involves itinerary planning. Many tourists still rely on manual methods such as notes, spreadsheets, or personal messaging applications to organize their travel schedules. These methods may not provide a convenient way to manage destinations, activities, and accommodations on a single platform. Consequently, tourists may experience difficulties in creating organized and flexible travel plans that suit their preferences and available time.

Tour guide services also play a vital role in providing visitors with a more informative and enjoyable travel experience. According to the Ormoc City tourism sector, there are 33

accredited tour guides that are members of the Ormoc Local Guides Association (OLGA). However, there is currently no centralized platform that efficiently manages tour guide availability and booking requests in a structured manner. This process may create delays in communication, difficulties in checking guide availability, and challenges in matching tourists with guides who possess relevant expertise.

In addition, local businesses such as accommodations, restaurants, souvenir shops, and other tourism-related establishments have limited opportunities to promote their services through a centralized tourism platform. While these businesses contribute significantly to the tourism ecosystem, many rely primarily on individual marketing efforts, making it difficult for tourists to discover services that could support their travel needs and improve their overall experience.

To address these concerns, the researchers propose the development of CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City. The proposed system will provide a centralized platform where tourists can browse destinations, create and manage itineraries, book accredited tour guides, and discover local tourism-related establishments. By integrating these services into a single platform, the system aims to improve accessibility to tourism information, streamline travel planning, support accredited tour guides and local businesses, and contribute to the promotion and sustainable development of tourism in Ormoc City.

Purpose and Description

The purpose of this project is to develop a unified digital tourism platform for Ormoc City that bridges the gap between tourists, tour guides, and local tourism-related services. The system addresses the fragmented nature of tourism information by providing a single platform where users can explore destinations, organize travel plans, and connect with accredited tour guides in a structured and efficient way.

The system supports tourists in making informed travel decisions by offering an organized directory of tourist destinations with essential details such as location, descriptions, and categories. Instead of relying on multiple sources of information, users can access everything within one platform.

A key feature of the system is its itinerary planning tool, which allows users to design flexible travel schedules based on selected destinations. This helps tourists manage their time effectively and ensures a more structured travel experience.

For tour guide services, the system introduces a simplified appointment mechanism where tourists can browse available accredited guides, review their profiles, and send booking requests.

The platform also extends beyond destinations and guides by incorporating local tourism-related businesses such as souvenir shops, accommodations, and dining establishments. This creates a more complete tourism ecosystem where visitors can easily discover services that enhance their travel experience.

Overall, the project introduces a more interactive and connected tourism model for Ormoc City, shifting from isolated information sources to an integrated system that supports planning, coordination, and discovery. It improves accessibility for tourists while also providing greater visibility for accredited tour guides, including members of the Ormoc Local Guides Association, and local tourism businesses.

Objective

General Objective:

To develop a centralized web and mobile-based tourism system for Ormoc City that integrates tourism information, itinerary planning, and tour guide booking features, improving accessibility and enhancing the overall travel planning experience of users.

Specific Objective:

- **To develop a system that allows users to explore and view complete information about tourist destinations in Ormoc City, including descriptions, locations, categories, and relevant details.**

    - This ensures that tourists can access accurate and updated destination information on a single platform without relying on multiple sources.

- **To implement a tour guide booking module that allows users to browse available accredited tour guides, view profiles, and check availability.**

    - This provides tourists with a convenient way to connect with suitable and reliable tour guides based on their travel needs.

- **To develop a booking and scheduling feature for tour guides based on their availability and preferred schedules.**

    - This allows users to conveniently reserve tour guides in advance, improving travel coordination and planning.

- **To develop an itinerary planning feature where users can create, customize, and manage their travel schedules based on selected destinations or tour guide suggestions.**

    - This enables tourists to organize their trips efficiently and choose structured travel plans for a better travel experience.

- **To provide a feature that allows tour guides to create and manage their own suggested itineraries for tourists.**

    - This allows experienced tour guides to share structured travel plans based on their expertise and guided tour knowledge.

- **To implement a local business integration feature that allows tourism-related establishments to submit their business information for inclusion in the system.**

    - This enables tourism administrators to review and approve submissions to ensure accurate and reliable listings of accommodations, and other services.

- **To develop an administrative dashboard for managing tourist destinations, tour guides, local businesses, bookings, and user activities.**

    - This enables administrators to monitor system operations and maintain organized and updated records.

- **To enable users to create and manage personal accounts for accessing tour guide booking and itinerary features.**

    - This allows users to securely store information, track their activities, and manage their travel plans effectively.

- **To implement a notification feature that informs users about booking status, itinerary updates, and system activities.**

    - This ensures users are updated in near real time regarding their travel-related actions and schedules.

- **To implement a tour guide registration and approval process that allows tour guides to submit their profiles for system inclusion.**

    - This enables administrators to verify and approve tour guides, ensuring quality and reliability of services.

- **To implement an in-app messaging feature that allows tourists and tour guides to communicate directly within the system.**

    - This enables both parties to discuss travel plans, booking details, schedules, meeting locations, and other tour-related concerns efficiently without relying on external communication platforms.

- To implement a QR code-based itinerary access feature that allows tourists to quickly view their assigned itinerary after a tour guide booking is confirmed.

    - This provides convenient and secure access to travel schedules and tour information through a scannable QR code.

Scope and Limitations

User levels

Administrator – Manages the overall system, including user accounts, tourist destinations, tour guide approvals, local business registrations, bookings, and system activities to ensure accurate and updated information.

Tour Guide – Registered and accredited users who provide guided tour services. They can manage profiles, set availability schedules, create suggested itineraries, and respond to booking requests.

Local Business Representative – Registered users who manage tourism-related establishments such as accommodations, restaurants, and souvenir shops. They can submit and update business information for administrator approval.

Tourist (User/Client) – Primary users of the system who can browse tourist destinations, create and manage itineraries, view tour guides, send booking requests, and access suggested itineraries.

Platform Accessibility

The system is accessible through both web and mobile platforms, allowing users to conveniently access tourism services using computers or mobile devices. Both platforms share a centralized database, ensuring real-time synchronization of bookings, updates, notifications, and user activities while providing a consistent user experience across devices.

Admin (Web-Based System)

Tourism Information Management – Manage all tourist destination data including descriptions, locations, images, categories, and travel details to ensure updated and accurate information for users.

Itinerary Content Oversight – Review and manage both user-generated and tour guide-generated itineraries to ensure they are valid, relevant, and aligned with available destinations in the system.

User Account Management – Handle registration, activation, deactivation, and monitoring of system users including tourists, tour guides, and local business representatives.

Tour Guide Verification – Validate tour guide applications, manage guide profiles, and ensure only accredited and approved guides are listed in the system, including their availability and specialization.

Tour Guide Booking Monitoring – Track and manage all booking requests made by tourists, including appointment status, guide assignment, and booking history records.

Local Business Integration Management – Review, verify, and approve submissions from local businesses such as accommodations, restaurants, and other tourism-related services before inclusion in the system.

System Activity Monitoring – Monitor overall system usage including user interactions, bookings, itinerary creation, and guide engagements to ensure proper system performance and control.

Notification Control – Manage system-generated notifications such as booking updates, approvals, itinerary changes, and important announcements for users and tour guides.

Local Business Representative (Web Based System)

Business Profile Management – Manage business information such as name, description, location, contact details, images, services offered, and emergency contact numbers to ensure accurate representation within the system.

Business Registration Submission – Submit application for inclusion in the system by providing complete business details for review and approval by the administrator.

Service Management – Update and maintain available services such as accommodations, food services, or tour-related offerings to keep information accurate and current.

Business Information Updates – Edit and update business details when necessary to ensure that tourists receive correct and up-to-date information.

Service Scheduling Management – Manage business schedules by setting operating hours, and available dates to help tourists plan visits and reservations efficiently.

Tour Guide (Web and Mobile Based System)

Profile Management – Allows tour guides to create and update their personal profiles including credentials, specialization, experience, and contact information to establish credibility within the system.

Availability and Schedule Management – Enables tour guides to set and update their availability, manage working schedules, and indicate dates when they are open for bookings.

Booking Request Management – Allows tour guides to receive, review, accept, or decline booking requests from tourists and manage appointment details efficiently.

Itinerary Creation – Provides tour guides with the ability to design and publish suggested itineraries based on their expertise, including destination sequences, time schedules, and recommended activities.

Tourism Content Reference – Gives tour guides access to approved tourist destination information within the system to help them create accurate and well-structured itineraries.

Appointment Coordination – Supports coordination between tour guides and tourists by managing booking confirmations and facilitating communication through shared contact information after approval.

Activity Monitoring – Allows tour guides to view their booking history, completed tours, and performance records within the system.

Suggested Itinerary Viewing and Downloading – Allows tourists to view pre-made itineraries created by tour guides and download them for offline use or personal travel reference.

Tourist (Web and Mobile Application)

Account Management – Allows tourists to register, log in, and manage their personal profiles, including basic information and travel preferences within the system.

Tourist Destination Browsing – Enables users to explore and view detailed information about tourist destinations such as descriptions, locations, categories, images, and travel details.

Search and Filtering – Provides tourists with tools to search for destinations and filter results based on categories, location, or type of attraction for easier navigation.

Itinerary Planning – Allows tourists to create, customize, and manage their own travel itineraries by selecting destinations and organizing them into a structured schedule.

Tour Guide Booking – Enables tourists to view accredited tour guides, check their profiles and availability, and send booking or appointment requests.

Tour Guide Itinerary Viewing – Allows tourists to view suggested itineraries created by tour guides and choose travel plans based on their preferences.

Local Business Discovery – Provides tourists with access to nearby or related tourism services such as accommodations, restaurants, and other establishments.

Booking History – Allows tourists to track and review their past and current tour guide bookings and itinerary activities.

Emergency Information Access – Allows tourists to view emergency contact numbers such as local police, hospitals, and tourism emergency hotlines for safety and assistance during their travel.

QR Code Itinerary Access – Allows tourists to receive and scan a QR code after booking confirmation to instantly view their assigned itinerary, tour schedule, and other travel details.

LIMITATIONS OF THE STUDY

- The system does not include advanced artificial intelligence-based travel recommendations, relying only on user selection and tour guide–created itineraries.

- The proposed system is limited to providing tourism information, itinerary planning, tour guide booking, and local business integration for Ormoc City only.

- The availability and accuracy of tour guide schedules depend on manual updates provided by the tour guides themselves, but blocks booked days automatically, and enforces a 30-minute buffer between tours and a same-day booking cutoff,

- GCash payment recording in demo mode. A live payment gateway such as PayMongo is outside the scope.

- The system does not include advanced artificial intelligence-based travel recommendations; itineraries can be auto-generated using rule-based logic.

Review of Related Literature/Studies/Systems

Related Literature

Recent studies have highlighted the growing importance of web-based tourism systems in improving the efficiency, accessibility, and overall experience of travelers. As tourism continues to shift toward digital platforms, researchers emphasize the role of centralized systems in simplifying destination discovery, tour guide services, itinerary planning, and tourism management. These systems address the limitations of traditional tourism processes, which often rely on fragmented information sources and manual coordination.

Digital Transformation in Tourism

The tourism industry has significantly evolved due to the rapid development of digital technology. According to Buhalis and Law (2008), information and communication technologies (ICT) have transformed the way tourists search, plan, and experience travel. Online platforms now serve as essential tools for accessing destination information, making travel decisions, and managing tourism services in a more convenient manner.

Similarly, Gretzel et al. (2015) emphasized that modern tourists increasingly rely on digital and mobile systems to support travel planning and decision-making. These systems provide integrated services such as destination information, itinerary planning, and service coordination, improving both efficiency and user experience.

In relation to the proposed system, these studies highlight the importance of integrating tourism services into a centralized platform, supporting the development of CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City.

Web-Based Tourism and Service Integration Systems

Web-based tourism systems are designed to centralize tourism information and services into a single platform. According to Law et al. (2018), centralized tourism systems improve accessibility and decision-making by organizing travel-related data such as destinations, services, and user activities in a structured manner.

In addition, Ramento et al. (2024) stated that web-based management systems enhance operational efficiency by automating manual processes such as bookings and scheduling. These systems reduce errors, improve data accuracy, and increase user satisfaction through real-time updates.

These findings support the proposed system, which integrates tourist destination information, tour guide hiring, and itinerary management into a unified platform for improved tourism service delivery.

Tour Guide Services in Tourism Systems

Tour guides play an essential role in enhancing tourist experiences by providing assistance, cultural interpretation, and local knowledge. According to Weiler and Black (2015), tour guides contribute significantly to tourist satisfaction and engagement by offering personalized and meaningful travel experiences.

Furthermore, integrating tour guide services into digital platforms improves accessibility and service efficiency. Digital tour guide systems allow tourists to easily view profiles, check availability, and book services in advance, reducing the need for manual coordination.

This supports the inclusion of a tour guide hiring module in the proposed system, enabling tourists in Ormoc City to connect directly with available and verified tour guides.

Itinerary Planning and Personalization in Tourism

Personalized itinerary planning is an important feature of modern tourism systems. According to Buhalis (2003), travelers prefer customized travel experiences that match their preferences, time, and budget.

In addition, Xiang et al. (2015) emphasized that personalization in tourism systems increases user engagement and satisfaction by providing tailored travel plans and recommendations.

This is relevant to the proposed system, which allows users to create personalized itineraries and view pre-made itineraries created by tour guides for structured travel planning.

Tourism Systems in the Philippine Context

In the Philippines, tourism plays a major role in local economic development. According to the Department of Tourism (DOT), digital platforms are increasingly being used to promote destinations and improve accessibility for tourists.

However, many local tourism systems still function mainly as informational websites and lack integrated features such as tour guide hiring, itinerary management, and booking services. This limitation creates a gap in providing a complete digital tourism experience.

This gap supports the development of a centralized system in Ormoc City that integrates tourism information, tour guide hiring, and itinerary planning into a single platform to improve efficiency and user experience.

Synthesis

Based on the reviewed literature, it is evident that digital transformation has significantly changed the tourism industry by improving how travelers access information, plan trips, and engage with tourism services. Studies emphasize that Information and Communication Technology (ICT) and mobile-based platforms have become essential tools in enhancing travel convenience, decision-making, and overall user experience.

The literature also highlights the importance of web-based tourism systems that centralize information and services into a single platform. These systems improve accessibility, efficiency, and accuracy by reducing manual processes and integrating essential tourism functions such as destination information management, booking services, and real-time updates.

Furthermore, studies show that tour guides play a vital role in enhancing tourist experiences by providing local knowledge, cultural interpretation, and personalized assistance. The integration of tour guide services into digital platforms allows tourists to conveniently access profiles, check availability, and book services, improving service coordination and user satisfaction.

Lastly, in the Philippine context, while digital platforms are increasingly used to promote tourism, many existing systems remain limited to information dissemination and lack integrated features such as tour guide hiring, booking systems, and itinerary management. This gap highlights the need for a more comprehensive and centralized tourism system.

Therefore, the reviewed literature supports the development of CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City, which aims to provide an integrated and efficient platform for tourism information, tour guide services, and itinerary planning in Ormoc City.

METHODOLOGY

Technical Background

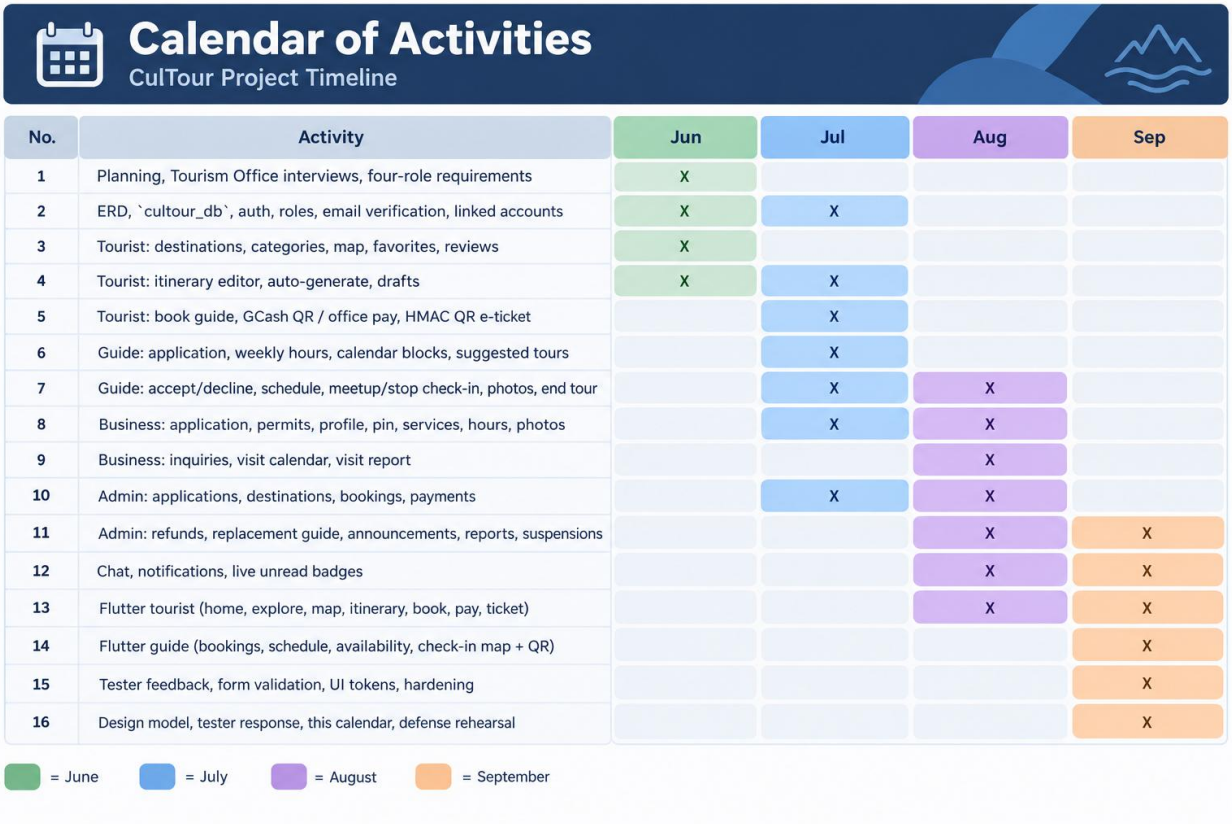
The research proponents gathered relevant information and studied the appropriate technologies, frameworks, and tools needed in developing the proposed system. These technologies will be used to ensure that the system is efficient, responsive, secure, and user-friendly across both web and mobile platforms, improving its overall performance and reliability.

The proposed project entitled CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City is designed for administrators, tourists, tour guides, and local business users. The system aims to centralize tourism services such as destination browsing, itinerary planning, tour guide booking, tour guide-generated itineraries, and local business listings into one platform to improve tourism accessibility and organization in Ormoc City.

In developing the system, the proponents will use front-end and back-end technologies, database systems, and development tools to ensure smooth system operation and proper data management. Each technology plays an important role in building the user interface, system functionality, and data processing.

The system will use HTML, CSS, JavaScript, Bootstrap, Tailwind CSS, and PHP for web development, with MySQL as the database. XAMPP and Apache will be used as the local server, Visual Studio Code as the code editor, and Flutter with Android Studio for mobile development.

Calendar of Activies



No.	Activity	Jun	Jul	Aug	Sep
1	Planning, Tourism Office interviews, four-role requirements	X			
2	ERD, `cultour_db`, auth, roles, email verification, linked accounts	X	X		
3	Tourist: destinations, categories, map, favorites, reviews	X			
4	Tourist: itinerary editor, auto-generate, drafts	X	X		
5	Tourist: book guide, GCash QR / office pay, HMAC QR e-ticket		X		
6	Guide: application, weekly hours, calendar blocks, suggested tours		X		
7	Guide: accept/decline, schedule, meetup/stop check-in, photos, end tour		X	X	
8	Business: application, permits, profile, pin, services, hours, photos		X	X	
9	Business: inquiries, visit calendar, visit report			X	
10	Admin: applications, destinations, bookings, payments		X	X	
11	Admin: refunds, replacement guide, announcements, reports, suspensions			X	X
12	Chat, notifications, live unread badges			X	X
13	Flutter tourist (home, explore, map, itinerary, book, pay, ticket)			X	X
14	Flutter guide (bookings, schedule, availability, check-in map + QR)				X
15	Tester feedback, form validation, UI tokens, hardening				X
16	Design model, tester response, this calendar, defense rehearsal				X

= June = July = August = September

Resources

The proponents will use the following tools and technologies in developing the system:

XAMPP – *will serve as the local development environment, providing Apache Server and MySQL for testing and system deployment during development.*

MySQL – *will be used as the database management system to store important data such as user accounts, bookings, itineraries, destination details, and transaction records.*

HTML (HyperText Markup Language) - *will be used to structure the web pages of the system.*

CSS (Cascading Style Sheets) - *will be used for styling and improving the visual design of the system.*

Bootstrap 5.3 – *will be used to ensure responsive design, allowing the system to adapt to different screen sizes such as desktop and mobile devices.*

JavaScript – *will be used to add interactivity to the system, including dynamic content updates, form validation, and user interface enhancements.*

PHP (Hypertext Preprocessor) – *will serve as the main server-side scripting language responsible for processing bookings, managing system logic, and communicating with the database.*

Apache Server – *included in XAMPP, will be used to host and run the web application locally during development and testing.*

Visual Studio Code – *will be used as the primary code editor for developing the system, offering features such as debugging, extensions, and version control support.*

Flutter (Dart) / Android Studio (Mobile Development) – *will be used if a mobile version of the system is developed, allowing cross-platform or native Android application development. fad*

Requirements Analysis

The proposed CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City is designed to improve tourism services in Ormoc City by involving key users such as tourists, tour

guides, administrators, and local business representatives. Each user plays an important role in ensuring that the system operates efficiently and provides a convenient and organized tourism experience.

Tourists can explore tourist destinations, view detailed information, create personalized itineraries, and browse suggested itineraries created by tour guides. They can also view available accredited tour guides and send booking or appointment requests through the system. Once a booking is confirmed, the system facilitates communication by providing contact information for both parties to coordinate through messaging.

Tour guides can manage their profiles, set availability schedules, and create pre-made itineraries based on their expertise. They can also review and respond to booking requests from tourists and manage their scheduled tours.

The administrator manages the overall system by maintaining tourist destination information, managing users and tour guides, reviewing local business registration requests, and monitoring booking activities.

Local business representatives can submit their business information such as accommodations, and food establishments for inclusion in the system, subject to administrator approval.

Overall, the system provides a centralized and interactive approach to tourism management by integrating destination information, itinerary planning, tour guide booking, and local business promotion into one platform. It improves accessibility, enhances coordination between users, and supports a more organized tourism experience in Ormoc City.

Requirements Documentation

This section presents the functional and non-functional requirements of the proposed CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business Integration for Ormoc City. The system is designed to provide a centralized platform that allows tourists to access tourism information, explore destinations, create itineraries, view tour guides, and send booking requests within Ormoc City.

The requirements were identified through observation, review of existing tourism systems, and interviews with relevant stakeholders such as the Ormoc City Tourism Office. The proponents also considered common issues experienced by tourists, including scattered tourism information, manual coordination with tour guides, and lack of organized trip planning tools.

Based on the information gathered, the system includes key functionalities such as tourist destination browsing, itinerary management, tour guide booking, user account management, and local business registration. Instead of payment processing, the system focuses on booking requests where tour guides and tourists are connected by sharing contact information for external communication.

These requirements serve as the foundation for the system's design and development, ensuring that it addresses the needs of tourists, tour guides, administrators, and local tourism stakeholders in Ormoc City.

1. FUNCTIONAL REQUIREMENTS

- FR-01 - The system shall allow tourists, tour guides, and business representatives to register an account with personal details, address, and contact information.
- FR-02 - The system shall allow users to log in with email and password, with an optional "remember me" session.
- FR-03 - The system shall allow users to reset a forgotten password through an emailed reset link.
- FR-03 - The system shall allow users to change their password and email address.

- FR-04 - The system shall require a password change when an account is flagged for one, for example an account for the admin created.
- FR-05 - The system shall allow one person to link several role accounts (for example Tourist and Tour Guide) and switch between them.
- FR-06 - The system shall allow users to view and update their profile, including photo and emergency contact person.
- FR-07 - The system shall allow suspended users to submit an appeal to the Tourism Office.
- FR-08 - The system shall allow users to log out and end their session.
- FR-09 - The system shall display tourist destinations with description, category, photos, location, services, operating hours, and emergency contact information.
- FR-10 - The system shall allow tourists to search for destinations and filter them by category.
- FR-11 - The system shall display destinations and businesses as pins on an interactive map.
- FR-12 - The system shall allow tourists to add and remove destinations and businesses from their favorites.
- FR-13 - The system shall allow tourists to rate and review destinations and businesses.
- FR-14 -The system shall display approved local businesses with their services, photos, operating hours, and contact details.
- FR-15 - The system shall allow tourists to send inquiries and messages to businesses.
- FR-16 - The system shall display published announcements and banners from the Tourism Office.
- FR-17 - The system shall allow tourists to add, edit, remove, and reorder destination and business stops, each with a visit date and time.
- FR-18 - The system shall allow tourists to set a meetup place and time for each day of the itinerary.
- FR-19 - The system shall auto-generate stops for a selected day using rule-based logic, leaving the other days unchanged.
- FR-20 - The system shall ask tourists to save the draft, discard it, or keep editing when leaving an itinerary with unsaved work.
- FR-21 - The system shall allow tourists to view and copy suggested itineraries created by tour guides.
- FR-22 - The system shall allow tourists to delete their own itineraries.
- FR-23 - The system shall list approved, accredited tour guides with profile, languages, experience, rate, rating, and review count.

- FR-24 - The system shall show which guides are available for the chosen itinerary dates and block guides who are unavailable.
- FR-25 - The system shall allow tourists to declare PWD or accessibility needs and upload a fitness certificate (up to 8 MB).
- FR-26 - The system shall generate a booking e-ticket with a signed QR code once the guide approves the booking.
- FR-27 - The system shall allow tourists to view and download the booking ticket as a PDF.
- FR-28 - The system shall allow tourists to cancel pending bookings. For a paid booking, it shall collect a GCash number or QR for the refund.
- FR-29 - When the original guide steps down, the system shall let tourists accept a replacement guide, choose another guide, move the remaining days, or request a refund.
- FR-30 - The system shall display booking history with status, payment status, refund status, and tour progress.
- FR-31 - The system shall allow tourists to rate their guide after a completed tour.
- FR-32 - The system shall allow tourists to chat with their tour guide inside the system.
- FR-33 - The system shall allow tour guides to apply with credentials, accreditation documents, languages, experience, and rate for admin approval.
- FR-34 - The system shall allow tour guides to set weekly availability and block specific calendar dates.
- FR-35 - The system shall automatically mark booked days as unavailable and enforce a 30-minute buffer between tours.
- FR-36 - The system shall notify tour guides of new booking requests and allow them to accept or decline once the booking is paid.
- FR-37 - The system shall require a reason, chosen from preset reasons, when a guide declines or cancels a booking.
- FR-38 - The system shall allow guides to cancel an approved tour before it starts, or withdraw from the remaining days of a multi-day tour.
- FR-39 - The system shall offer a withdrawn booking to other available guides, each with a 15-minute window to accept.

- FR-40 - The system shall allow guides to scan and verify a tourist's booking QR ticket at the meetup.
- FR-41 - The system shall allow guides to check in and record each stop, checked in with a photo, or skip.
- FR-42 - The system shall allow guides to end each tour day and mark the tour completed. Tours whose dates have passed are completed automatically.
- FR-43 - The system shall allow guides to chat with tourists inside the system.
- FR-44 - The system shall allow businesses to register with business details and supporting documents for admin approval.
- FR-45 - The system shall allow businesses to manage their profile, including name, type, description, location on the map, contacts, and photos.
- FR-46 - The system shall allow businesses to manage their services and operating hours.
- FR-47 - The system shall allow businesses to receive and reply to tourist inquiries and messages.
- FR-48 - The system shall display a visit calendar showing how many people are booked to visit the business each day.
- FR-49 - The system shall display a visit report with daily and monthly totals of expected and completed visits.
- FR-50 - The system shall display engagement statistics such as views, favorites, and reviews.
- FR-51 - The system shall allow admins to add, edit, publish, and archive destinations with photos, services, hours, category, and map location.
- FR-52 - The system shall allow admins to manage categories and business types.
- FR-53 - The system shall allow admins to review, approve, or reject tour guides and business applications.
- FR-54 - The system shall allow admins to create, view, activate, and suspend tourist, guide, and business accounts.
- FR-55 - The system shall allow admins to review suspension appeals and reactivate accounts.
- FR-56 - The system shall allow admins to monitor all bookings, view booking details, verify payments, and assign a replacement guide.

- FR-57 - The system shall allow admins to monitor tours in progress and their stop check-ins.
- FR-58 - The system shall allow admins to review and manage tourist and guide itineraries.
- FR-59 - The system shall allow admins to post announcements and banners.
- The system shall allow admins to handle content reports.
- FR-60 - The system shall generate reports on bookings, users, destinations, and tourism activity.
- FR-61 - The system shall record user and admin actions in an activity log.
- FR-62 - The system shall send in-app notifications for registration approval, booking requests, approvals, declines, cancellations, payments, replacements, refunds, and announcements.
- FR-63 - The system shall send email notifications for key booking and account events.
- FR-64 - The system shall update badges, booking lists, and chat automatically without a manual page refresh.
- FR-65 - The system shall notify guides before an upcoming tour.

2. NON- FUNCTIONAL REQUIREMENTS

- NFR-01 - The system shall store passwords only as bcrypt hashes.
- NFR-02 - The system shall require passwords of at least 5 characters with at least one letter and one number.
- NFR-03 - The system shall lock login for 15 minutes after 5 failed attempts per email

- NFR-04 - The system shall protect every web form with a CSRF token.
- NFR-05 - The system shall use prepared statements for all database queries to prevent SQL injection.
- NFR-06 - The system shall restrict every page and API endpoint to its authorized role.
- NFR-07 - The system shall end idle web sessions after 8 hours
- NFR-08 - The system shall accept only allowed image and document file types for uploads, up to 8 MB each.
- NFR-09 - The system shall block direct web access to configuration, source code, database, and storage folders.
- NFR-10 - The system shall sign QR ticket links with a token so tickets cannot be forged.
- NFR-11 - The system shall refresh live data
- NFR-12 - The system shall provide a responsive web interface that works on desktop, tablet, and mobile screens.
- NFR-13 - The system shall use the same colors, layout, and terms on web and mobile.
- NFR-14 - The system shall show clear validation and confirmation messages for every user action.
- NFR-15 - The system shall ask for confirmation before destructive actions such as delete, cancel, or discard.
- NFR-16 - The system shall offer first-time tutorials for key features such as the app and itinerary planning.
- NFR-17 - The system shall use one shared database for web and mobile, so data stays consistent across devices.
- NFR-18 - The system shall enforce relationships between tables with foreign keys.
- NFR-19 - The system shall prevent double-booking guides on overlapping dates and times.
- NFR-20 - The web system shall run on current versions of Chrome, Edge, and Firefox.
- NFR-21 - The mobile app shall run on Android and Chrome through Flutter.
- NFR-22 - Web pages and the mobile API shall use the same shared logic, so one change applies to both platforms.
- NFR-23 - The system shall support simultaneous use by all four user roles on web and mobile.

- NFR-24 - The system shall be available whenever the host server (Apache and MySQL) is running.

3. Technical Requirements

3.1 Front-End

• The system shall be developed using HTML, CSS, and JavaScript, HTML shall be used to define the structure of web pages, CSS shall be used for styling and layout design,

JavaScript shall be used to add interactivity such as dropdown menus, form validation, and calendar selection.

3.2 Back-End

• The system shall use a MySQL database for storing and managing data such as user accounts, tourism information, bookings, payments, and itineraries.

- The system shall use PHP to handle dynamic content such as displaying destinations, booking information, and itinerary data.

• The system shall support online server communication between the front-end and back-end for real-time data processing.

3.3 Security

• The system shall use secure password hashing, CSRF protection, five-attempt login lockout, an eight-hour session timeout, account suspension and appeals, and file-upload validation.

3.4 Tools

• The system shall be developed using Visual Studio Code and XAMPP as the main development tools.

• The system design and interface prototypes shall be created using Figma and Canva for UI/UX planning.

Design of Software, System, Product, and/or Processes

This section presents the design of the proposed CulTour: A Web and Mobile-Based System for Destination Discovery, Itinerary Planning, Licensed Tour Guide Booking and Local Business

Integration for Ormoc City. The system is developed following a standard web and mobile application architecture to ensure efficiency, usability, scalability, and maintainability.

The system follows a client-server architecture, where the client-side handles user interaction through web and mobile interfaces, while the server-side manages system logic, data processing, authentication, and database operations. This structure enables smooth and efficient communication between users and the system through internet-based platforms.

The system is organized into key functional modules to ensure clear structure and functionality. These include tourist destination management, tour guide management, itinerary management, tour guide booking, local business integration, user management, and notification services. Each module works together to provide a centralized tourism platform for Ormoc City.

In the itinerary module, tourists can create their own personalized travel plans based on available destinations, while tour guides can also create suggested itineraries based on their expertise. Users can browse and adopt these itineraries to support their travel planning**.** It also allows users to view suggested itineraries created by tour guides and download them in PDF format for offline access and travel convenience.

The system process flow starts with user registration and login, followed by browsing tourist destinations and tour guides, creating or selecting itineraries, and sending tour guide booking requests. Once a booking is confirmed, the system facilitates communication by sharing contact details for external coordination between tourists and tour guides.

The database design uses a relational database structure to store and manage system data such as user accounts, tour guide profiles, tourist destinations, booking records, itineraries, and local business information. Relationships between tables are properly established to ensure data consistency, integrity, and efficient data retrieval.

After a booking is confirmed, the system generates a unique QR code linked to the booking ticket. Tourists can scan the QR code using their mobile devices to view itinerary details, destinations, schedules, and tour information without manually searching through the system.

Overall, the system design ensures that all components work together to provide a centralized, efficient, and user-friendly tourism management system that improves travel planning and tourism coordination in Ormoc City.

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APPENDICES

APPENDIX A. RESOURCE PERSONS

DAVE DELACRUZ – TOURISM OFFICE ADMINISTRATIVE AIDE

REX FRANZE ESTONILO – MEMBER OF ORMOC LOCAL GUIDES ASSOCIATION

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

Steve Clarence D. Amores



Brgy. Balugo, Albuera Leyte

Steveclarence02@gmail.com

09665320037

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Ormoc
Tertiary	2022-2022	San Lorenzo Ruiz College
Vocational/Technical	2019-2021	San Lorenzo Ruiz College
High School	2015-2019	DGBZMSF
Elementary	2009-2015	Tabgas Elementary School

AFFILIATION

Inclusive Dates	Name of Organization	Position
2022-Present	IT CLUB	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Programming Languages- Python, Java	Basic	January 2024
Database- MySql	Basic	January 2025
Esports Player(MLBB)	Amateur	March 2020

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, Workshop
2016	Dare- Drug Abuse Resistance Education
2025	Cyber Security Awareness- STI COLLEGE OF ORMOC
2025	Blockchain Tech Workshop(PAYTACA)-STI COLLEGE OF ORMOC

Curriculum Vitae of

Czar Aian Judilla

Brgy. San Jose, Ormoc City
czaraianj@gmail.com
09380350682



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-present	STI College Ormoc
Vocational/Technical	2020-2022	Genaro B. Lureñana NHS
High School	2016-2020	Genaro B. Lureñana NHS
Elementary	2010-2016	San Jose Elementary School

AFFILIATION

Inclusive Dates	Name of Organization	Position
2022-present	IT CLUB	Member
2020-2022	Cycling Team	Member

SKILLS

SKILLS	Level of Competency	Data Acquired
Programming Language – Java, C#, python, JavaScript	Basic	2024-2026
Data Base – MySQL, SQL, SQLite	Basic	2024
Cycling	Basic	2020-2022

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
2023	SAP Business one – STI College Ormoc
2025	Block Chain Tech Workshop – STI College Ormoc
2026	Bridging Academe & Industry Through Research – STI College Ormoc

Curriculum Vitae of

Nick Andre N. Suson

Brgy. Kadaohan, Ormoc City

Nickandresuson@gmail.com

09154388206



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/Institution
Tertiary	2023 - Present	STI College of Ormoc
Tertiary	2022 - 2022	University of San Carlos
Vocational/ Technical	2020 - 2022	Western Leyte College of Ormoc
High School	2016 - 2020	Valencia National High School
Elementary	2010 - 2016	Valencia Central School

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2023 - Present	IT Club	Member
2025 - Present	Ormocanon Road Pacers	Member
2016 - 2020	Boy scout of The Philippines	Assistant Senior Crew Leader
2016 - 2018	Red Cross Youth	Member
2018 - 2019	War on Waste	Member

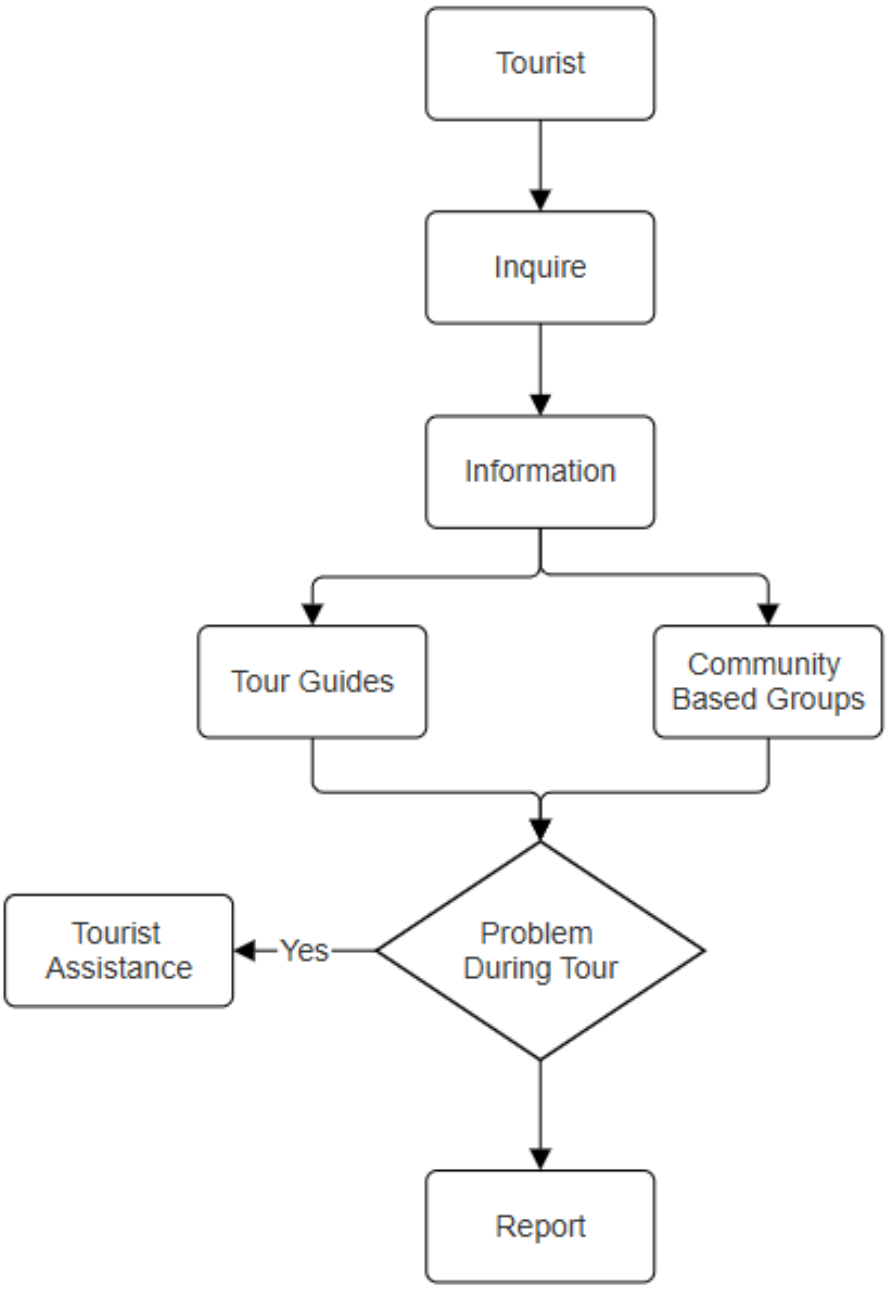
SKILLS	Level of Competency	Date Acquired
Programming Languages – Java, C#, Python, PHP, JavaScript	Basic	Jan 2023, Nov 2024, Sept 2023, Feb 2026, Feb 2026 Respectively
Database – MySQL, Sql, SQLite	Basic	Jan 2023
Running	Basic	2025

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

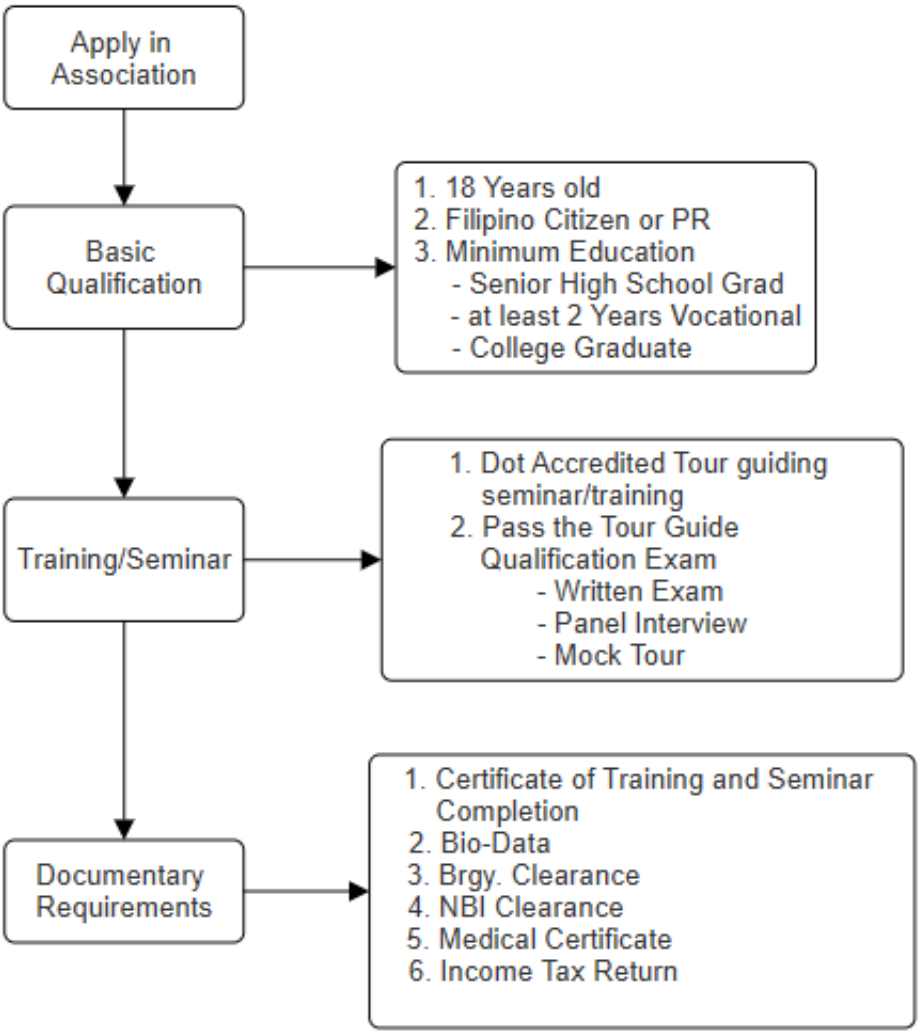
Inclusive Dates	Title of Training, Seminar, or Workshop
2016	Dare – Drug Abuse Resistance and Education
2023	SAP Business One - STI College Ormoc
2025	Mind Matters – Cultivating Mental Wellness

APPENDICES DESIGN MODEL

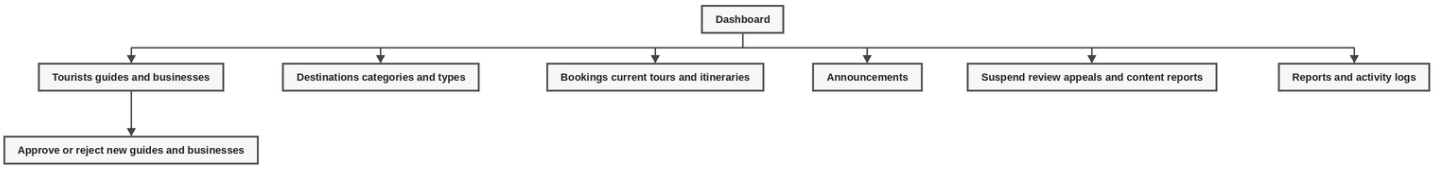
Manual Design Model



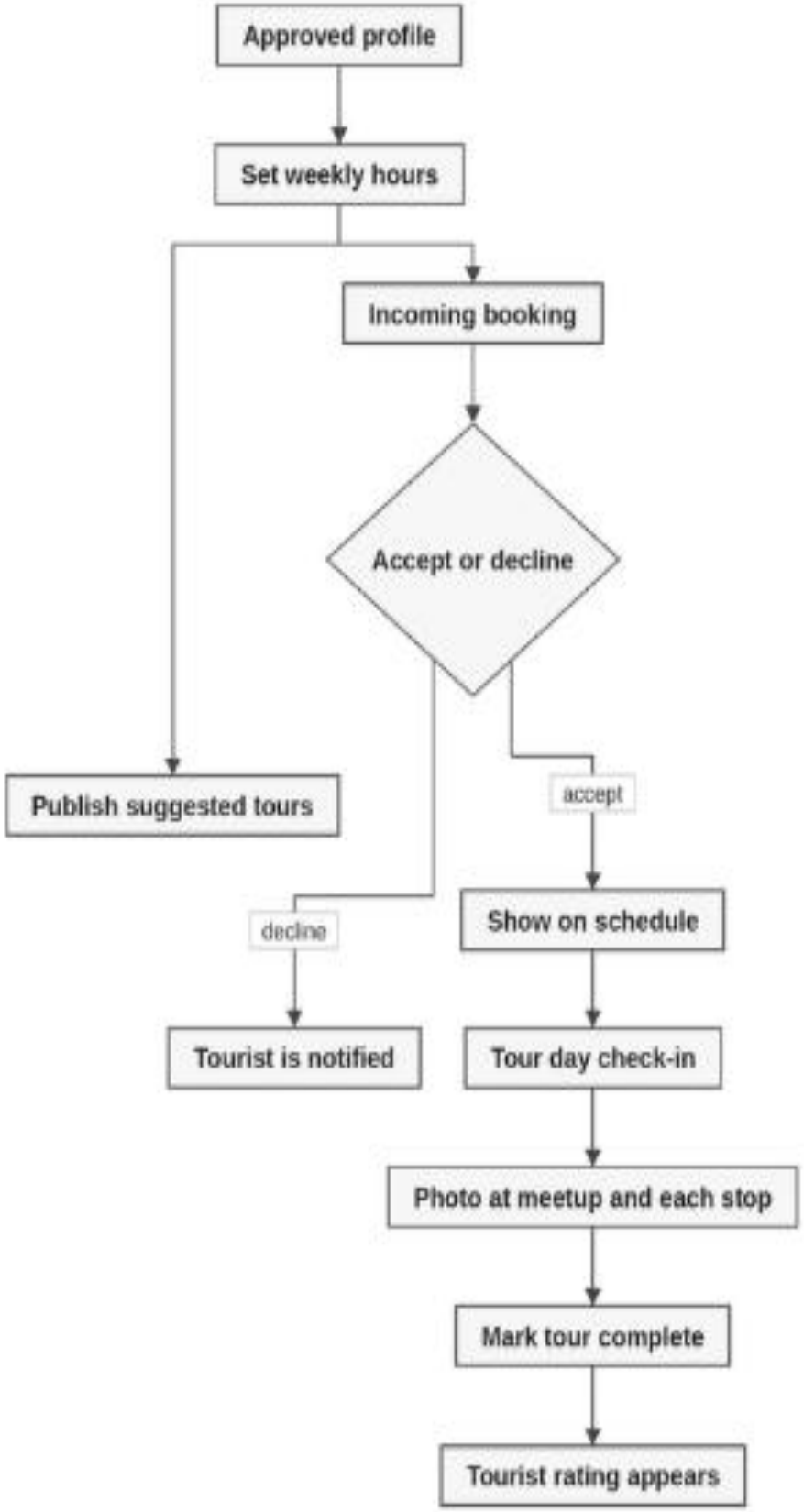
Tour Guide Accreditation Design model



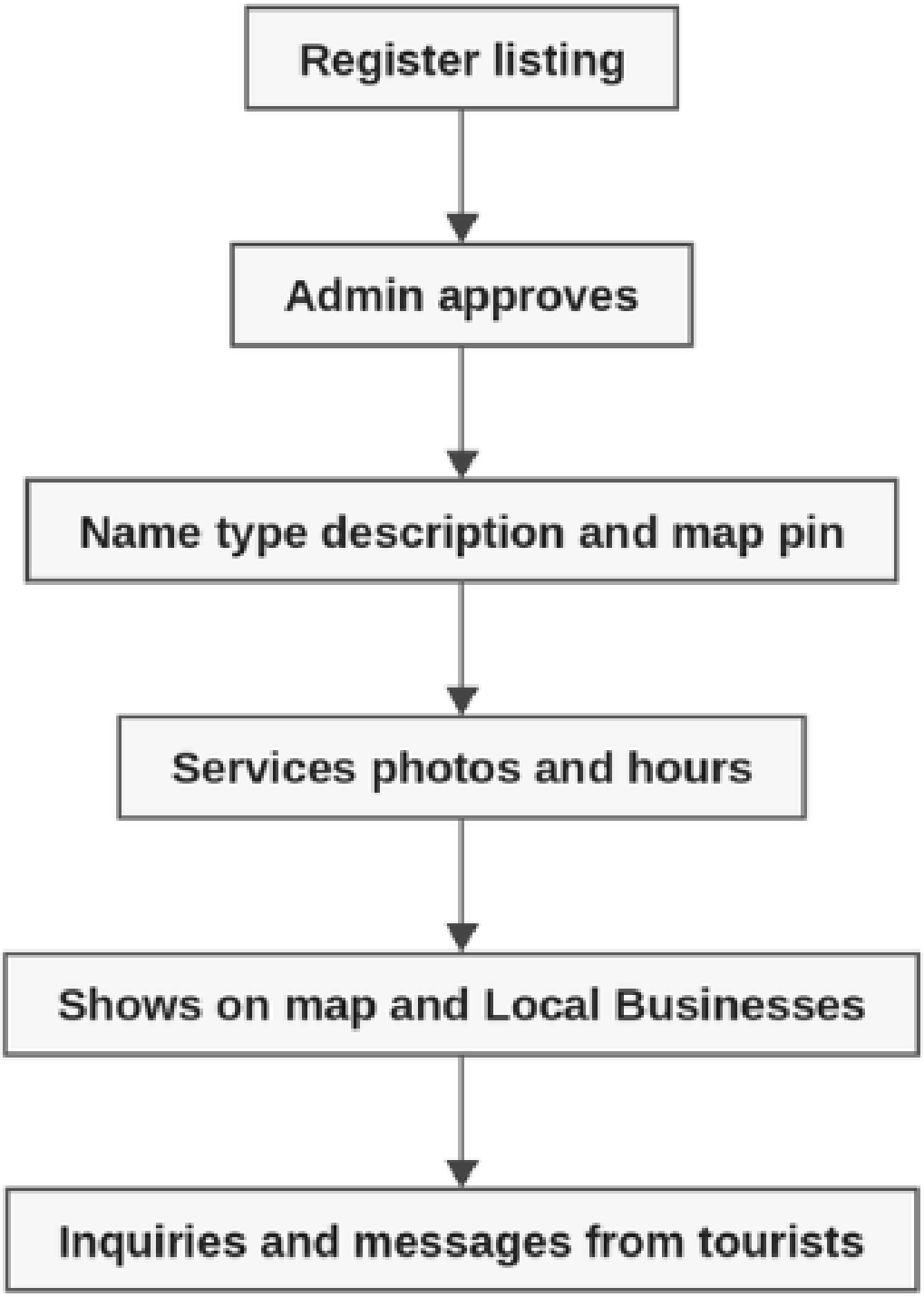
PROPOSED ADMIN DESIGN MODEL


Dashboard
Tourists guides and businesses
Destinations categories and types
Bookings current tours and itineraries
Announcements
Suspend review appeals and content reports
Reports and activity logs
Approve or reject new guides and businesses

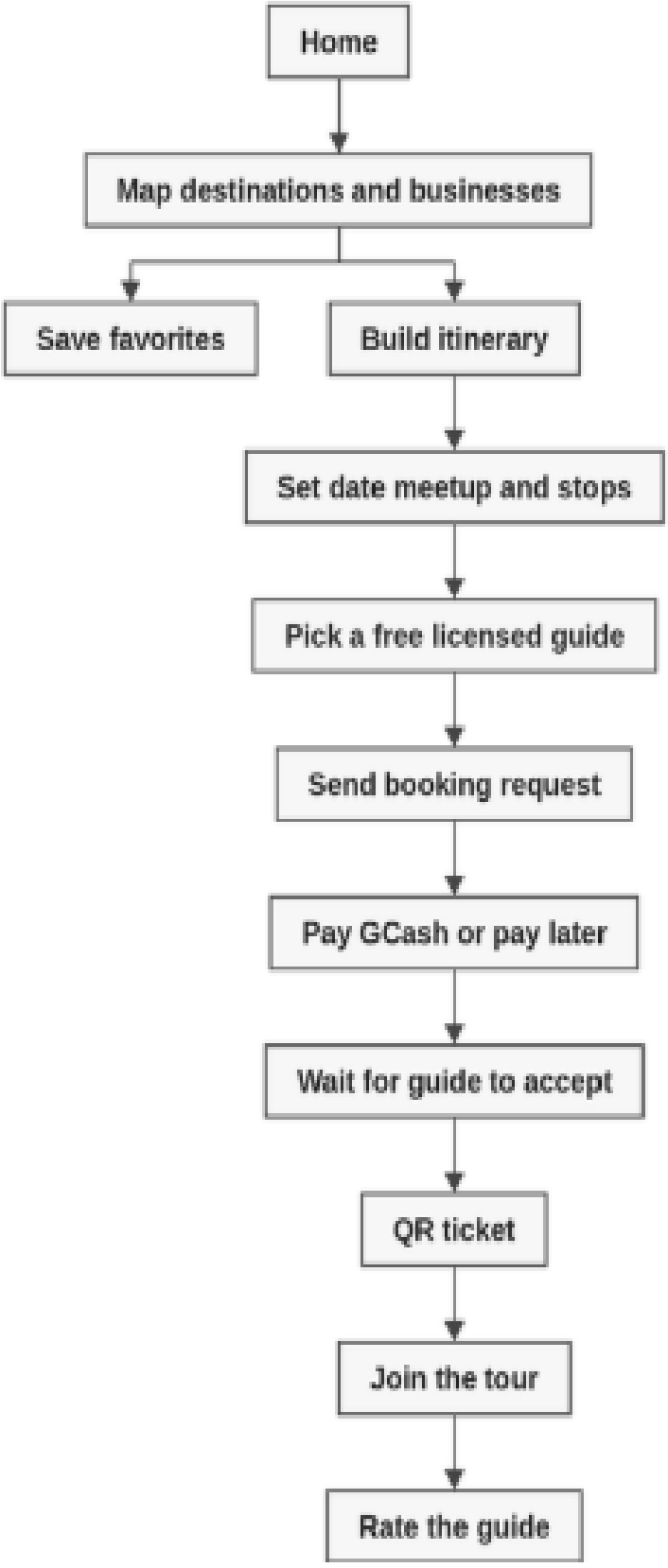
PROPOSED TOUR GUIDE DESIGN MODEL


Approved profile
Set weekly hours
Incoming booking
Accept or decline
Publish suggested tours
accept
decline
Show on schedule
Tourist is notified
Tour day check-in
Photo at meetup and each stop
Mark tour complete
Tourist rating appears

PROPOSED BUSINESS DESIGN MODEL


Register listing
Admin approves
Name type description and map pin
Services photos and hours
Shows on map and Local Businesses
Inquiries and messages from tourists

PROPOSED TOURIST DESIGN MODEL


Home
Map destinations and businesses
Save favorites
Build itinerary
Set date meetup and stops
Pick a free licensed guide
Send booking request
Pay GCash or pay later
Wait for guide to accept
QR ticket
Join the tour
Rate the guide